

Pandas

Pandas is a software library used for data manipulation and analysis. (Python library)

Pandas main datastructure

① Data frame.

② Series.

eg →

cgpa	iq	Placement
8.0	80	0
7.0	90	0
7.1	100	1
6.1	200	1

→ Data frame

Series.

Importing data

Firstly we need to get the csv file of the data.
• csv → comma separated values.

We can upload the csv file using upload button in our jupyter notebook.

`import pandas as pd` → to import lib in our code.

`pd.read_csv(demo.csv)` used to import data.

Important functions & attributes

`data.head()`

df object → only to see preview of the data.

→ Give first 5 rows or we can see as much rows as wanted.

`data.tail()` → get preview from the last of the data frame.

`data.shape` → tells us the shape of a given dataset.

`data.info()` gives us information about the dataframe.

`data.describe()` → Works only on numerical features & give info about them.

Fetching column and rows out of a dataframe.

in series we have index & values.

`df['winner']` → to fetch column from dataframe.

`df[['winner', 'team1', 'team2']]` → multiple column.

`data.iloc[0:1]` → used to fetch a row from df.
↳ we can fetch row & columns both.

↓
`df.iloc[0:100, [0, 1, 2]]`

→ filtering data on a given condition.

we can filter out the data as we need.
example →

```
mask1 = df['winner'] == 'RCB'
mask2 = df['season'] >= 2010
S = df[mask1 & mask2]
Print(S)
```

→ (value_counts) → used to get frequency of elements in a particular series.

Plot function

• `Plot(kind = "_____")`
↳ type of graph we need

we need to import matplotlib lib to visualize this.

→ Series operations

Series only have two things index and values and if we apply any function on series we get a series.

series.index series.values

 → used to get indexes and values.

→ we can also apply head & tail on panda series also.

→ If two series have same indexes then we can perform arithmetical operations on them.

→ sort values function

This function is used to sort the values of a given series or a dataframe:

```
df['win-by-runs'].sort_values()
```

```
sort_values(ascending=False)
```

↳ used to sort in ascending order.

→ we can use them on dataframes as well.

```
df.sort_values('city')
```

→ we can use it for multiple columns inside a data frame.

```
df.sort_values(['city', 'date'])
```

→ drop_duplicates() function.

This function is used to remove duplicate values from the data from multiple columns.

```
df.drop_duplicates(['city', 'season'])
```

→ (groupby function)

we can create groups as we need of our data.

```
data.groupby('sector')
```

→ `sector.first()` used to create a df of first occurrence of the group.

→ `sector.last()` → to create for last occurrence.

→ `sector.groups` → tells all the groups and tells us the members of that group.

→ `sector.get_group("...")`

↳ this is used to get dataframe of a particular group.

→ `sector.sum()` → to get sum of all numerical values.

• these are some common commands on groups.

• `isin()`

This is used to check that a list of objects are in the dataframe or not.

Merge function

→ This can be used to merge multiple dataframes.

`data1.merge(data2, left_on=..., right_on=...)`

↳ used to merge two columns.

→ Pivot table.

table made on a dataframe which makes decision making easier.

`Pivot-table(index, columns, values, aggfunc)`

→ Correlation function.

→ gives us the relation between numerical values of the dataframe.

→ (Rename function)

used to rename columns inside a particular data frame.

```
data.rename(columns = {'City': 'location'})
```

→ (set_index) or (reset_index)

↓
used to set index of a particular column as index column.

↓
used to reset to original pandas.

→ Handling missing values.

(remove missing values)

```
→ data.dropna(how = 'all', subset = [ ], axis = _)
```

↓
how will it remove

↓
which column to check for

↓
column or row.

→ (filling missing values)

→ `df["..."].fillna(method="ffill")`

↳ specific column.

we can put method
or a specific value
or string that we
want.